

Notes: Artavanis et al. (RFS, 2025)

Cross-Subsidization of Bad Credit in a Lending Crisis

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1 Big picture

- **Question.** During a lending crisis, do banks knowingly charge stronger borrowers above-cost rates and use these rents to subsidize weaker borrowers with low or even negative markups?
- **Setting.** The paper studies the large-corporate loan portfolio of a major Greek systemic bank during the Greek financial crisis. The special feature of the data is that each loan comes with both the *actual rate* and the bank's internally computed *break-even rate* (BE rate), which is essentially the bank's marginal cost of making the loan.
- **Core measurement contribution.** Most zombie-lending papers must infer whether a loan is subsidized by comparing observed rates to indirect benchmarks such as prime rates, ratings, or interest-coverage ratios. This paper directly observes the benchmark and therefore directly observes the markup:

$$\text{Markup}_{imt} = \text{Actual Rate}_{imt} - \text{BE Rate}_{imt}.$$

- **Main theoretical mechanism.** In normal times, loan pricing reflects funding costs and default risk. In crisis times, collateral values are depressed and banks cannot cheaply raise outside capital. Then low-profitability borrowers are not liquidated immediately because waiting has option value, while high-profitability borrowers can be charged positive markups because rival banks would have to liquidate existing loans to poach them. That creates de facto cross-subsidization.
- **Three main predictions.**
 - *Limited pass-through:* actual rates respond less than one-for-one to BE rates.
 - *Crisis sensitivity:* pass-through becomes weaker when the crisis deepens and stronger when the crisis abates.
 - *Asymmetry:* pass-through is stronger for safer, more profitable borrowers and weaker for riskier borrowers whose rates are driven more by ability to pay.
- **Main baseline result.** The cross-sectional pass-through coefficient from BE rates to actual rates is about 0.30 in the baseline regressions. A 1 percentage point increase in the BE rate raises the actual interest rate by only about 30 basis points.
- **Direct evidence of subsidization.** Roughly 14.6% of loans are issued at rates at least 100 basis points below BE, with an average negative markup of about 4.02%. At the same time, many safer borrowers pay large positive markups.
- **Crisis-timing result.** Around May–July 2015, when Greece neared sovereign default, the pass-through coefficient fell to about zero. After the July 13 third memorandum, pass-through jumped by about 0.46 to 0.49.
- **Cross-sectional asymmetry.** For below-median initial BE borrowers, pass-through is about 0.43. For above-median initial BE borrowers, it is essentially zero once time and firm fixed effects are included.
- **Main interpretation.** Riskier borrowers are priced according to what they can afford to pay; safer borrowers are priced according to a limit-pricing rule that extracts rents from them.
- **Credibility results.** The paper shows the pattern is not just loan managers ignoring the new pricing rule, not foresight about future borrower quality, not narrow focus on regulatory capital charges, and not an artifact of standard indirect zombie-loan definitions.

- **Economic takeaway.** A banking crisis can distort the cross-section of credit prices even when banks know the true marginal cost of each loan. Healthy firms may be overcharged precisely when weak firms are being kept alive.

2 Data and measurement (key objects)

2.1 Why this setting is unusually informative

- The paper exploits a Greek regulatory reform after the 2012 recapitalization of systemic banks. Monitoring institutions required banks to adopt formulaic, transparent, and data-driven loan-pricing models.
- Each bank had to compute the BE rate of every loan using exogenous inputs such as risk assessments, collateral values, and balance-sheet variables, with uniform rules across comparable loans.
- Loans priced below BE required a special internal approval and a written justification.

Interpretation. Most papers only observe the final interest rate. This paper also observes the bank's own cost benchmark, which sharply reduces the usual joint-hypothesis problem in zombie-lending work.

2.2 Main sample

- The data come from a large Greek systemic bank that passed the 2014 EBA stress tests.
- The sample covers the bank's *large corporate portfolio*, that is, firms above a size threshold.
- The main sample runs from January 2015 to June 2017 for new loan contracts, with an extension back to January 2013 used for pre-guideline comparisons.
- The core sample contains 1,625 loan contracts and 150 borrowers.
- The sample includes term loans and credit lines and covers some of the largest and most established firms in Greece.

Interpretation. The paper is not about micro or retail credit. It is about large corporate lending by a major bank at the peak of a systemic crisis.

2.3 How the break-even rate is constructed

The BE rate is the lowest interest rate that makes the loan non-loss-making for the bank. It combines a fixed and a variable component. The main moving part is credit risk.

- **Expected-loss component.** This depends on the loan balance, collateral value, and a formula-based probability of default.
- **Regulatory-capital component.** This captures the capital charge associated with unexpected losses, using Basel-style models and a hurdle rate.

- **Funding and operating cost.** These are built into the fixed part of the pricing model.

Interpretation. The BE rate is not a market quote; it is the bank’s internal marginal-cost measure. The crucial point is that it is computed in a disciplined, rule-based way from outside risk inputs and accounting data, not managerial discretion.

2.4 Key variables

- **Actual rate:** the contractual interest rate on the loan.
- **Break-even rate (BE):** the formulaic cost benchmark.
- **Markup:**

$$\text{Markup}_{imt} = \text{Actual Rate}_{imt} - \text{BE Rate}_{imt}.$$

Positive markup means the bank charges above cost; negative markup means the loan is subsidized relative to the bank’s own pricing model.

- **Borrower controls:** operating return on assets (OROA), deposits, total assets, and liabilities-to-assets.
- **Loan controls:** maturity, collateral indicator, and loan amount.
- **Rating:** an externally developed borrower credit model input, mapped one-to-one to probability of default and updated over time.

2.5 Summary statistics

Table 1 reports the main facts.

- Mean actual rate: 5.38%.
- Mean BE rate: 4.76%.
- Mean markup: 0.62 percentage points, or 62 basis points.
- Markup dispersion is large: the 5th percentile is -3.57% , while the 95th percentile is 4.46% .
- Mean loan amount: 3.62 million euros.
- Mean maturity: 0.62 years, so the sample is tilted toward short-term loans that are rolled over often.
- About 85% of loans are collateralized.
- At the borrower level, mean OROA is 4.59%, mean deposits are 18.88 million euros, mean total assets are 237.88 million euros, and liabilities-to-assets average 50.87%.

Interpretation. The average loan is priced above BE, but the cross-section is wide enough to contain both large positive markups and meaningfully negative markups. That dispersion is the empirical object the paper wants to explain.

2.6 Why firm size matters for external validity

Figure 2 compares the log-asset distribution of the 150 sample firms with the universe of 23,835 Greek firms in Amadeus. The sample is concentrated among the largest firms in the country.

Interpretation. This is useful because the sample consists of economically important corporate borrowers, but it also means the paper's conclusions are most directly about large-firm lending, not small-business banking.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic environment

The paper sets up a dynamic model with two states: *postcrisis* and *crisis*. A firm's cash flow can be in a high-profitability regime H or a low-profitability regime L . Transition rates between these regimes are borrower-specific.

- In the **postcrisis** state, banks can raise capital freely and collateral values are high.
- In the **crisis** state, collateral values are depressed and banks have limited or no access to fresh capital.

Interpretation. The whole point of the model is to show why a lending crisis changes not just the level of rates, but the *mapping* from marginal cost to final price.

3.2 Postcrisis benchmark

When the economy is in the postcrisis state, bank debt values satisfy

$$R_H + p_L(\max\{V_L, C^*\} - V_H) = r^*V_H, \quad (1)$$

$$R_L + p_H(V_H - V_L) = r^*V_L. \quad (2)$$

Competition implies $V_H = 1$, so the bank breaks even on high-profitability borrowers, and under the parameter restrictions in the paper the low-profitability borrower is liquidated in normal times. This yields

$$R_H = r^* + p_L(1 - C^*). \quad (3)$$

Interpretation. This is the standard benchmark: the loan rate equals funding cost plus expected default loss. No cross-subsidization appears in normal times.

3.3 Crisis state and limit pricing

During the crisis, collateral falls from C^* to C , and banks must finance new lending internally. The debt-pricing equations become

$$R_{H,c} + p_L(V_{L,c} - V_{H,c}) + \rho(1 - V_{H,c}) = rV_{H,c}, \quad (4)$$

$$R_{L,c} + p_H(V_{H,c} - V_{L,c}) + \rho(C^* - V_{L,c}) = rV_{L,c}. \quad (5)$$

Because rivals would have to liquidate existing loans to poach an H -type borrower, the no-poaching condition becomes $V_{H,c} = D/C \geq 1$ rather than $V_H = 1$.

Under the paper's parameter restrictions,

$$R_{L,c} = \pi_L, \quad (6)$$

$$R_{H,c} = r + (r + \rho) \left(\frac{D}{C} - 1 \right) + p_L \left(\frac{D}{C} - V_{L,c} \right), \quad (7)$$

with

$$V_{L,c} = \frac{\pi_L + p_H \frac{D}{C} + \rho C^*}{r + \rho + p_H} < \frac{D}{C}. \quad (8)$$

Interpretation. The key asymmetry is transparent here. Low-profitability borrowers are priced by *ability to pay*: the bank extracts current cash flow and waits rather than liquidating. High-profitability borrowers are priced by a *limit-pricing* condition because competitor banks are constrained.

3.4 Limited pass-through prediction

The model implies that the cross-sectional difference in rates between L and H borrowers is smaller than the corresponding difference in frictionless marginal cost. Formally, the paper writes

$$\mathbb{E}_i(R_{L,c}^i) = \mathbb{E}_i(R_{H,c}^i) + \beta \rho(1 - C^*), \quad \beta < 1. \quad (9)$$

Interpretation. This is the formal statement of *imperfect pass-through*. If BE is the empirical counterpart of marginal cost, the coefficient of actual rate on BE should be below one.

3.5 Baseline empirical pass-through regression

The main estimating equation is

$$\text{Actual Rate}_{imt} = \alpha_t + \alpha_s + \beta \text{BE Rate}_{imt} + X_{it}\delta + X_m\eta + \varepsilon_{imt}, \quad (10)$$

where i indexes borrowers, m loans, and t months.

- α_t are time fixed effects.
- α_s are industry fixed effects.
- X_{it} are borrower controls.
- X_m are loan controls.

Interpretation. The coefficient β is the empirical pass-through from the bank's internal marginal cost to the final contractual interest rate. A value below one means the bank compresses pricing differences across borrowers.

3.6 Lagged-rate specification

Because managers may rely on recent borrower-specific pricing information, the paper also estimates

$$\text{Actual Rate}_{imt} = \alpha_t + \alpha_s + \beta \text{BE Rate}_{imt} + \rho \text{Actual Rate}_{i,t-1} + X_{it}\delta + X_m\eta + \varepsilon_{imt}. \quad (11)$$

Interpretation. If the main result disappears after controlling for the borrower's recent rate, then BE may just be proxying for stale information. If it survives, the BE rate contains independent pricing information.

3.7 Crisis-abatement specification

To test whether pass-through strengthened after the July 13, 2015 memorandum, the paper estimates

$$\text{Actual Rate}_{imt} = \alpha_t + \alpha_i + \beta \text{BE Rate}_{imt} + \delta D_{post} \times \text{BE Rate}_{imt} + \varepsilon_{imt}, \quad (12)$$

where $D_{post} = 1$ after July 13, 2015.

Interpretation. The coefficient on $D_{post} \times \text{BE Rate}$ measures how much pass-through rose when the crisis suddenly abated.

3.8 Asymmetric borrower-level pass-through

The paper then re-estimates the pass-through equation separately for borrowers with below-median and above-median initial BE rates, including both time and firm fixed effects.

Interpretation. The model says safer borrowers in the high-profitability regime should still face pricing tied to marginal cost, while weaker borrowers in the low-profitability regime should face rates tied mostly to current ability to pay. The split sample is the empirical implementation of that idea.

3.9 Pre-post guideline adjustment equations

To test whether the new pricing rules mattered in practice, the paper studies how 2015 actual rates moved relative to 2014 actual rates:

$$\Delta \text{Actual Rate}_{i,2015} = \alpha_s + \beta (\text{BE}_{i,2015} - \text{Actual Rate}_{i,2014}) + \Delta X_{i,2015}\delta + \varepsilon_i, \quad (13)$$

and also an inertia version,

$$\text{Actual Rate}_{i,2015} = \alpha_s + \beta \text{BE}_{i,2015} + \rho \text{Actual Rate}_{i,2014} + X_{i,2015}\delta + \varepsilon_i. \quad (14)$$

Interpretation. These equations ask whether the introduction of the pricing model in early 2015 actually shifted loan pricing in the direction implied by the newly computed BE rates.

3.10 Future-rating test

To test whether low markups simply reflect managers' superior private information about future borrower quality, the paper estimates

$$\text{Rating}_{i,t+h} = \alpha_t + \alpha_s + \phi \text{Markup}_{i,t} + \theta \text{Rating}_{i,t} + \varepsilon_{i,t}, \quad (15)$$

for horizons up to 60 months.

Interpretation. If managers give low rates to borrowers because they privately know those borrowers will improve, then today's markup should forecast future rating upgrades. No such forecasting power would undercut the superior-information story.

4 Identification and instruments (what makes the estimates credible)

4.1 Main source of credibility: direct observation of BE rates

The paper's most important identification advantage is not a classical external instrument. It is the fact that the researcher directly observes the bank's own formulaic marginal-cost benchmark for each loan.

Interpretation. This sharply narrows the inferential problem. The paper does not need to guess which loans were subsidized using noisy proxies; it can see the subsidy directly as a negative markup.

4.2 Why reverse causality is weak

The actual rate is chosen by the loan manager, but the BE rate is generated by the rule-based pricing model from exogenous risk assessments, collateral, and balance-sheet information.

Interpretation. A manager can react to BE, but the contractual rate itself does not feed back into the BE formula in a way that would mechanically generate the main results.

4.3 Fixed effects and common shocks

The baseline regressions include time fixed effects and, in many specifications, industry fixed effects. The crisis-abatement tests also include firm fixed effects.

Interpretation. These controls absorb market-wide shifts in rates, industry-specific shocks, and time-invariant borrower heterogeneity. The remaining variation is closer to loan-level pricing compression rather than macro interest-rate trends.

4.4 Policy timing around July 2015

The May–July 2015 episode is a sharp macro event. Failed negotiations with creditors, the June 27 referendum announcement, bank closures, and then the July 13 third memorandum created a sudden worsening and then easing of the crisis.

Interpretation. This is not a perfect quasi-experiment, but it is a powerful within-sample event that lines up closely with the theory's prediction that crisis severity should affect pass-through.

4.5 Could managers simply ignore the new BE rate?

Section 7.1 shows they did not. When the pricing rules were introduced, low-initial-BE borrowers on average received a rate cut while high-initial-BE borrowers received a rate increase. In the formal regressions, a 1% discrepancy between the 2015 BE rate and the 2014 actual rate translated into roughly a 40 basis point adjustment.

Interpretation. If the BE rate were just a bureaucratic sideshow, pricing would not move this strongly when the rule was introduced.

4.6 Could managers have superior private information?

The future-rating regressions show that current markups do not positively predict later rating improvements. The estimated markup coefficients are mostly negative and insignificant.

Interpretation. This weakens the idea that managers price below BE because they privately know those borrowers are about to improve. The evidence instead fits the paper's simpler pricing-distortion story.

4.7 Could managers be responding only to the regulatory-capital part of BE?

The paper isolates the regulatory-charge component of the BE rate and adds it separately in an appendix specification. Its coefficient is near zero and insignificant, while the BE coefficient remains intact.

Interpretation. Managers are not just passing through the most salient regulatory piece of cost and ignoring the rest.

4.8 Do expensive borrowers leave the bank?

If positive markups simply reflected temporary market power that pushed good firms away, one would expect high-markup borrowers to shrink their relationship with the bank over time. The paper finds no such pattern.

Interpretation. This supports the limit-pricing interpretation: the bank sets rates high, but not so high that profitable borrowers leave.

4.9 Why standard zombie-loan proxies are imperfect

The paper compares three labels for subsidized credit: actual below BE, actual below a prime-rate benchmark, and interest-coverage ratio below one. The prime-rate rule identifies only a narrow subset of subsidized loans, while the ICR rule yields both false positives and false negatives.

Interpretation. Direct observation of negative markups is genuinely informative; it is not redundant with usual zombie-lending proxies.

4.10 What the paper can and cannot distinguish

- The model is stylized and maps profitability regimes into reduced-form borrower categories.
- The paper identifies pricing compression and its crisis dependence very clearly.
- It is less definitive about the exact structural microfoundation among all possible crisis-friction stories, though limit pricing, depressed collateral, and ability-to-pay pricing fit the facts well.

5 Tables and figures

5.1 Figure 2: sample firms versus all Greek firms

Read: The figure plots the distribution of log assets for the 150 borrower firms in the sample against the universe of Greek firms in Amadeus. The sample sits far to the right of the full-firm distribution.

Economic message. The paper studies very large firms, not the representative Greek borrower. This makes the sample especially relevant for aggregate credit allocation, but also means the results are about large-corporate loan pricing.

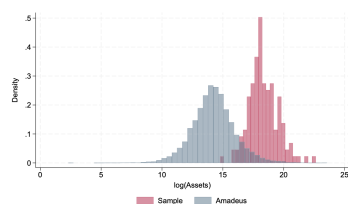


Figure 2
Sample distribution of log(Assets) versus all Greek firms
This figure reports the distribution of log(Assets) for the 150 sample firms (red) and all 23,835 Greek firms in the 2014 Amadeus data (blue).

Table 1 Summary statistics								
A. Loan-level variables								
	Count	Mean	SD	P5	P25	P50	P75	P95
Actual rate (%)	1625	5.38	1.71	2.47	4.65	5.48	6.10	7.85
Break-even rate (%)	1625	4.76	3.34	1.97	2.36	4.44	5.57	10.98
Markup (%)	1625	0.62	2.73	-3.57	-0.02	0.49	2.28	4.46
Loan amount (million €)	1625	3.62	14.35	0.04	0.20	0.90	2.50	10.04
Maturity (years)	1581	0.62	1.51	0.08	0.17	0.25	0.50	1.99
Collateralized (%)	1625	85.17	35.55	0.00	100.00	100.00	100.00	100.00
B. Borrower-level variables								
	Mean	SD	P5	P25	P50	P75	P95	
OROA (%)	4.59	5.97	-5.14	1.16	4.65	7.41	15.40	
Deposits (million €)	18.88	81.72	0.38	1.48	4.18	9.18	58.11	
Total assets (million €)	237.88	655.06	15.86	43.02	78.35	175.72	832.10	
Liabilities/Assets (%)	50.87	22.59	13.59	34.13	49.82	64.64	92.07	

This table reports the summary statistics. Panel A reports the loan-level variables for the 1625 loans opened from 2015 and until the end of our sample period (2017m6) described in section 2.4, and panel B reports the summary statistics for borrower-level variables for the 150 borrowers in our main sample. Detailed variable definitions are provided in Table A1 in the Internet Appendix.

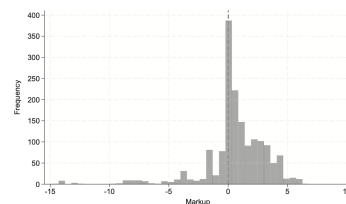


Figure 3
Histogram of markups
Histogram of the distribution of markups, defined as the difference between actual and break-even rates.

5.2 Table 1 and Figure 3: descriptive facts and markup dispersion

Read: Table 1 shows that average actual rates exceed average BE rates by 62 basis points, but the markup distribution is wide. Figure 3 visualizes the markup histogram with both negative and positive tails.

Economic message. The average loan is not subsidized, but many individual loans are. Cross-subsidization is therefore a cross-sectional phenomenon, not just a low average lending margin.

5.3 Figure 4 and Table 2: the baseline limited pass-through result

Read: Figure 4 plots actual against BE rates for all new loans. Many points lie above the 45-degree line, but a meaningful set lies below it. The fitted line is much flatter than the 45-degree line. Table

2 quantifies this: the pass-through coefficient is 0.299 to 0.310 in the baseline columns and stays around 0.28–0.29 as controls are added. Once firm fixed effects are included, the coefficient drops to 0.095 and becomes imprecise.

Economic message. Across borrowers, banks do not pass marginal cost through one-for-one. Safer borrowers pay large positive markups; riskier borrowers often receive compressed or negative markups. The firm-fixed-effects result foreshadows the later asymmetry result: most within-firm pass-through comes from safer borrowers.

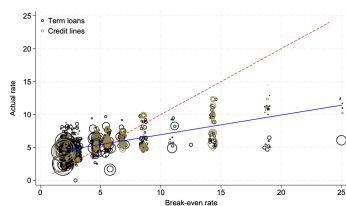


Figure 4
Actual rate and break-even rate
This figure presents the scatterplot of actual and break-even interest rates for new loans from 2015 to June 2017. The red dashed line corresponds to the 45° line. The blue solid line represents the OLS fitted line. The size of the circle corresponds to the relative size of the account. Black circles correspond to term loans, and brown circles correspond to credit lines.

	(1)	(2)	(3)	(4)	(5)	(6)
Break-even rate	0.299*** (0.033)	0.310*** (0.031)	0.285*** (0.035)	0.292*** (0.039)	0.280*** (0.038)	0.095 (0.097)
ROA			-0.016 (0.019)	-0.017 (0.019)	-0.026* (0.016)	
Deposits			-0.076 (0.088)	-0.055 (0.091)	-0.15*** (0.060)	
Assets			0.082 (0.132)	0.068 (0.138)	0.268*** (0.101)	
Liabilities/Assets			0.004 (0.006)	0.003 (0.006)	0.002 (0.006)	
Maturity			-0.064 (0.064)	0.006 (0.063)	-0.002 (0.062)	
Collateral			0.242 (0.194)	0.001 (0.163)	-0.245 (0.274)	
Loan amount			-0.034 (0.044)	-0.062 (0.040)	-0.023 (0.018)	
Observations	1,625	1,625	1,371	1,329	1,486	1,549
R ²	.339	.389	.390	.386	.446	.601
Quarter FE	No	Yes	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	No	No
Firm FE	No	No	No	No	No	Yes

This table reports the baseline estimates of Equation (18), where columns 1–5 incrementally add covariates. Column 6 reports the results from augmenting Equation (18) with firm fixed effects. The dependent variable is the actual interest rate of each loan (in all columns). Deposits and Assets are in natural logarithms. Standard errors, which are clustered by borrower, are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

	(1)	(2)	(3)	(4)	(5)	(6)
Break-even rate	0.186*** (0.042)	0.199*** (0.042)	0.154*** (0.042)	0.162*** (0.046)	0.197*** (0.039)	0.079 (0.108)
Actual rate _{t-1}	0.567*** (0.093)	0.556*** (0.091)	0.580*** (0.072)	0.583*** (0.074)	0.538*** (0.059)	0.525*** (0.106)
ROA			-0.012 (0.015)	-0.010 (0.016)	-0.013 (0.013)	
Deposits			0.053 (0.078)	0.066 (0.079)	-0.030 (0.060)	
Assets			0.063 (0.098)	0.063 (0.106)	0.277*** (0.082)	
Liabilities/Assets			0.002 (0.005)	0.002 (0.005)	0.001 (0.005)	
Maturity			-0.007 (0.058)	0.001 (0.053)	-0.011 (0.074)	
Collateral			0.156 (0.152)	-0.051 (0.152)	-0.249 (0.284)	
Loan amount			-0.020 (0.046)	-0.058 (0.040)	0.010 (0.020)	
Observations	1,423	1,423	1,389	1,364	1,331	1,376
R ²	.447	.488	.490	.498	.530	.614
Quarter FE	No	Yes	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	No	No
Firm FE	No	No	No	No	No	Yes

This table augments Equation (18) with a lagged actual rate. Specifically, Actual rate_{t-1} is calculated as the average interest rate of the borrower's loans in the most recent previous month. Columns 1–5 incrementally add the covariates of the specification, and column 6 includes firm fixed effects. Deposits and Assets are in natural logarithms. Standard errors, which are clustered by borrower, are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.4 Table 3: lagged-rate robustness

Read: After controlling for the borrower's lagged actual rate, the BE coefficient remains about 0.18 to 0.20 in the baseline columns, while the lagged-rate coefficient is about 0.54 to 0.58.

Economic message. There is inertia in loan pricing, but BE still matters. The limited pass-through finding is not just a reflection of sticky borrower-specific rates.

5.5 Figure 5 and Table 4: pass-through and the severity of the crisis

Read: Figure 5 plots weekly and monthly pass-through coefficients around the peak of the Greek crisis. They fall to about zero during the deep-crisis window and jump above 0.5 right after July 13, 2015. Table 4 shows the formal interaction estimate: $D_{post} \times BE$ is between 0.461 and 0.491 across specifications.

Economic message. The crisis matters for pricing. When financial conditions are at their worst, rates become almost disconnected from marginal cost. When the crisis eases, the bank restores more normal pass-through.

5.6 Table 5: asymmetric pass-through

Read: Splitting borrowers by initial BE rate, the paper finds coefficients of 0.434 and 0.428 for below-median-BE borrowers, but only 0.041 and 0.044 for above-median-BE borrowers once firm and time fixed effects are included.

Economic message. This is one of the sharpest pieces of evidence in the paper. Safer borrowers

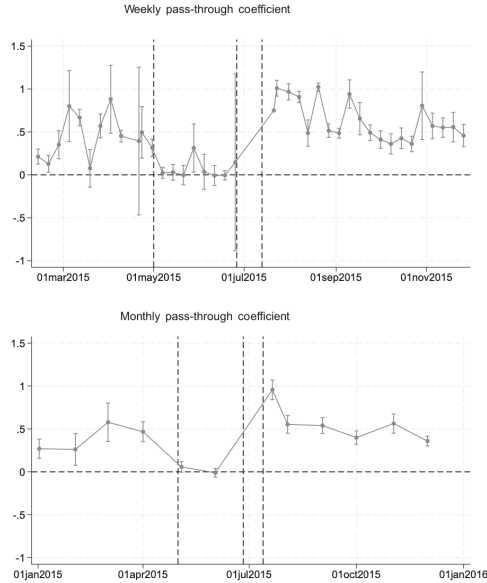


Figure 5
Time-series of pass-through coefficients around the peak of the crisis
Time series of pass-through coefficients from Equation (19). Top Plot: Weekly pass-through coefficient (with confidence bands). Bottom Plot: Monthly pass-through coefficient (with confidence bands). The three vertical dashed lines in the two figures correspond to (a) May 1 (the expiration of the deadline to conclude the technical negotiations between Greece and its official creditors), (b) June 26 (the announcement of a snap referendum), and (c) July 13 (the surprising agreement on a third memorandum). We refer to the period between the first two dotted lines as the “deep crisis” period, and the period after the third dotted line as the “crisis abatement” period.

Table 4
Pass-through coefficients before and after the third memorandum

	(1)	(2)	(3)
Break-even rate	0.018 (0.025)	0.003 (0.025)	0.128 (0.303)
$D_{post} \times \text{Break-even rate}$	0.461*** (0.035)	0.465*** (0.033)	0.491*** (0.025)
Observations	336	320	290
R^2	.731	.750	.835
Month FE	Yes	Yes	Yes
Industry FE	No	Yes	Yes
Firm FE	No	No	Yes

This table reports the results of estimating the Equation (20). The dependent variable is the actual rate of each loan in all columns. The sample is 2015m5-2015m12, that is, the sample encompasses the “deep crisis” period of May and June 2015 as well as the “crisis abatement” period (the 6 months after the signing of the third memorandum on July 13, 2015). D_{post} is an indicator variable taking the value 1 if the loan is signed after July 13, 2015. Standard errors, which are clustered by borrower, are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

are priced with meaningful cost pass-through; riskier borrowers are priced almost entirely outside that logic. That is exactly what the crisis model predicts.

Table 5
Asymmetric pass-through

	Below-median BE		Above-median BE	
	(1)	(2)	(3)	(4)
Break-even rate	0.434** (0.190)	0.428** (0.181)	0.041 (0.093)	0.044 (0.092)
Maturity		0.083* (0.041)		-0.082 (0.077)
Collateral		-0.271 (0.246)		0.052 (0.231)
Loan amount		-0.027 (0.033)		-0.023 (0.016)
Observations	544	532	883	863
R^2	.480	.497	.411	.432
Quarter FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

The dependent variable is the actual rate of each loan in all columns. Columns 1 and 2 (columns 3 and 4) include only borrowers with an initial BE rate in 2015 that is below (above) the median initial BE rate of the sample. Standard errors, which are clustered by borrower, are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

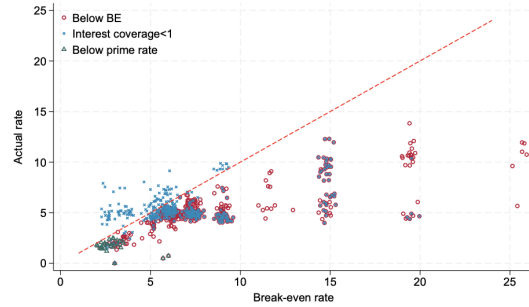


Figure 6
Alternative definitions of subsidization

This figure reports three alternative classification schemes for identifying subsidized loans. The green triangle points correspond to loans that receive a below-prime interest rate, defined as the minimum BE given to highest credit-rating firms in a month. The blue X-shaped points represent the loans for firms with interest coverage ratio below one for the previous 2 years. The red circle points are loans with interest rate below the BE (observations below the 45° line).

5.7 Figure 6: alternative definitions of subsidization

Read: The figure overlays three concepts of a subsidized loan: below-BE loans, below-prime-rate loans, and firms with interest coverage ratio below one. Below-prime loans are all below BE, but they are only a small subset of all subsidized loans. ICR-based labeling overlaps with subsidized loans only noisily.

Economic message. Traditional zombie-loan proxies contain some signal, but they are either too conservative or too noisy. Seeing the BE rate directly materially improves identification.

5.8 Figure 7 and Table 6: introduction of the new pricing guidelines

Read: Figure 7 shows that borrowers with low initial BE rates received average rate cuts from 2014 to 2015, medium-BE borrowers saw little change, and high-BE borrowers received rate hikes. Table 6 quantifies this: the coefficient on $BE_{2015} - AR_{2014}$ is about 0.40 in first differences, and the level specification gives a similar estimate around 0.38.

Economic message. Loan managers did respond to the newly introduced pricing model. So the main limited-pass-through result cannot be dismissed as pure noncompliance with the pricing framework.

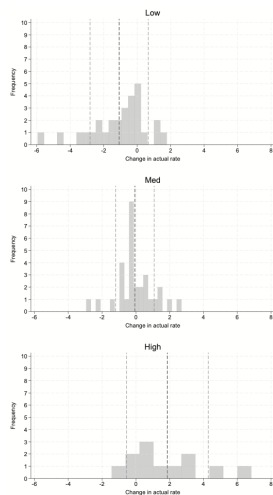


Figure 7
Actual rate changes by initial BE rate
These figures report the distribution of actual rate changes by initial BE rate. The sample is divided into three terciles, according to the initially reported BE rate ("low," "medium," "high") of each corporate borrower in 2015. Each plot shows a histogram of the changes to a borrower's actual rates from 2014 to 2015 for the corresponding tercile of initial BE rate. The black, vertical dashed lines represent the mean of each distribution, and the vertical, gray dashed lines represent the one-standard-deviation range around the mean.

Table 6
Introduction of the new pricing guidelines and actual rate changes

	A. First difference		
	AR ₂₀₁₅ - AR ₂₀₁₄		
	(1)	(2)	(3)
Break-even ₂₀₁₅ - AR ₂₀₁₄	0.403*** (0.053)	0.397*** (0.055)	0.409*** (0.060)
Δ OROA ₂₀₁₅		-0.055 (0.035)	-0.048 (0.060)
Δ Deposits ₂₀₁₅		0.287* (0.166)	0.263 (0.173)
Δ Assets ₂₀₁₅		-0.344 (0.990)	-0.676 (1.094)
Δ Liabilities/Assets ₂₀₁₅		-0.006 (0.008)	-0.003 (0.009)
Observations	70	65	63
R ²	0.27	0.65	0.66
Industry FE	No	No	Yes
	B. Levels		
	Actual rate ₂₀₁₅		
	(1)	(2)	(3)
Break-even ₂₀₁₅	0.379*** (0.033)	0.372*** (0.039)	0.377*** (0.042)
Actual rate ₂₀₁₄	0.051 (0.152)	0.089 (0.158)	0.108 (0.171)
OROA		-0.007 (0.023)	-0.005 (0.022)
Deposits		0.023 (0.143)	-0.000 (0.146)
Assets		0.127 (0.154)	0.137 (0.174)
Liabilities/Assets		0.008 (0.006)	0.007 (0.007)
Observations	70	67	65
R ²	0.25	0.44	0.63
Industry FE	No	No	Yes

This table reports the estimates of Equation (21) in panel A and Equation (22) in panel B. The dependent variable in panel A is the change in the borrower's average interest rate from 2014 to 2015. The dependent variable in panel B is the difference between the borrower's average 2015 BE rate and their average 2014 interest rate. Deposits and Assets are in natural logarithms. In panel A, OROA, Deposits, Assets, and Liabilities/Assets are in first differences. Robust standard errors are reported in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table 7
Future credit ratings

	A. All borrowers				
	(1) h=12	(2) h=24	(3) h=36	(4) h=48	(5) h=60
Dif	-0.093 (0.086)	-0.125 (0.090)	-0.175 (0.109)	-0.192* (0.106)	-0.268** (0.132)
Rating	0.695*** (0.094)	0.653*** (0.104)	0.487*** (0.118)	0.456*** (0.087)	0.348*** (0.101)
Observations	744	704	686	673	664
R ²	0.485	0.444	0.293	0.281	0.218
Quarter FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	No
	B. Negative Dif borrowers				
	(1) h=12	(2) h=24	(3) h=36	(4) h=48	(5) h=60
Dif	-0.077 (0.138)	-0.105 (0.107)	0.060 (0.124)	-0.007 (0.127)	0.124 (0.151)
Rating	0.746*** (0.168)	0.523*** (0.113)	0.604*** (0.124)	0.512*** (0.170)	0.511** (0.201)
Observations	173	148	143	142	141
R ²	0.76	0.625	0.584	0.509	0.329
Quarter FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	No
	C. Positive Dif borrowers				
	(1) h=12	(2) h=24	(3) h=36	(4) h=48	(5) h=60
Dif	0.111 (0.143)	0.111 (0.134)	0.116 (0.213)	0.029 (0.186)	-0.243* (0.146)
Rating	0.659*** (0.108)	0.628*** (0.121)	0.402*** (0.143)	0.392*** (0.100)	0.280** (0.121)
Observations	571	556	543	531	523
R ²	0.370	0.346	0.177	0.163	0.127
Quarter FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	No

The dependent variable in all columns is a borrower's rating at time $t+h$, where h is measured in months. Dif _{t} is the average difference of the actual minus the BE rate across all loans of borrower i in month t , that is, the markup in month t . Rating _{t} is the rating of borrower i at time t . Panel A shows the results for all observations. Panel B only includes observations, where the (initial) Dif _{t} is negative, and panel C only includes observations with (initially) positive Dif _{t} . A lower (higher) value of the variable "Rating" corresponds to better (worse) credit quality. Standard errors are clustered by borrower, and reported in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

5.9 Table 7: future credit ratings

Read: Regressing future borrower ratings on today's markup and current rating, the paper finds markup coefficients that are mostly negative and statistically insignificant across 12- to 60-month horizons, for all borrowers and separately for negative- and positive-markup subsamples.

Economic message. Low markups do not predict future borrower improvement. This weakens the superior-information story and strengthens the interpretation that the pricing pattern reflects deliberate cross-subsidization.

6 Short summary

- The paper studies a Greek bank during the sovereign-debt crisis using rare data that include both actual loan rates and internally computed break-even rates.
- The theoretical model combines depressed collateral values, limited access to outside capital, and limit pricing.

- In crisis times, weak borrowers are not liquidated because waiting has option value; strong borrowers can be charged positive markups because rivals are capital constrained.
- Empirically, the pass-through from BE to actual rates is far below one, about 0.30 in the baseline.
- A nontrivial share of loans is priced below BE, giving direct evidence of subsidized lending.
- Pass-through collapses during the peak of the Greek crisis and rises sharply after the July 2015 memorandum.
- Safer borrowers exhibit meaningful pass-through, around 0.43, while riskier borrowers exhibit essentially none.
- The paper rules out several alternatives: managers ignoring BE, managers pricing on superior private information, and managers passing through only regulatory cost.
- Standard indirect measures of zombie lending are informative but noisy relative to directly observed markups.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a distressed banking system can knowingly compress the cross-section of loan prices. Stronger borrowers are charged above-cost rates, while weaker borrowers receive smaller markups and sometimes outright subsidies relative to the bank's own break-even benchmark.

7.2 Economic intuition

The crisis changes the economics of both liquidation and competition. Depressed collateral values make immediate liquidation unattractive for weak borrowers, so the bank waits and charges what those borrowers can pay. At the same time, impaired access to outside funding weakens competition for strong borrowers, allowing incumbent banks to extract rents through limit pricing.

7.3 Takeaway

The paper gives clean evidence that credit misallocation during crises can operate through *pricing*, not just through quantities. The ability to observe break-even rates directly shows that cross-subsidization is deliberate and conditional on the bank's information set, not merely the by-product of coarse risk categories or hidden borrower heterogeneity.